Experiment No. 1

Date: / /

Aim: To determine the concentration in terms of molarity of $KMnO_4$ solution by titrating it aganist standard solution of oxalic acid.

Theory: The substances available in the state of high purity are used to prepare standard solutions by dissolving a fixed/definite mass in definite volume of distilled water. Oxalic acid is a primary standard substance. Its molar mass is 126 g/mol.

A. Preparation of standard solution of oxalic acid

Molar mass of oxalic acid ($H_2C_2O_4.2H_2O$) is 126 g/mol

For 1000 mL 1 M oxalic acid solution required mass is 126 g of oxalic acid

Hence for 100 mL of 0.1 M oxalic acid the required mass is

$$= \frac{126 \times 100 \times 0.1}{1000 \times 1}$$

$= 1.26$ g of oxalic acid

Apparatus: 100 mL standard flask, balance, watch glass, beaker, glass rod, etc.

Chemicals: Oxalic acid, distilled water.

Procedure:

1. Weigh accurately 1.26 g oxalic acid on watch glass.
2. Transfer the weighed oxalic acid to a beaker and wash the watch glass with distilled water and transfer washings to the beaker. Add little distilled water to dissolve it by stirring.
3. Transfer the solution of oxalic acid from beaker to 100 mL standard flask. Wash beaker twice with water and transfer washings to the 100 mL standard flask. Dilute the solution up to the mark on standard flask to make volume 100 mL.

B. Determination of molarity of $KMnO_4$ solution using standard solution of oxalic acid.

Apparatus: Burette, pipette, conical flask, burner, water bath, etc.

Chemicals: $KMnO_4$ solution, 0.1 M oxalic acid, 2M H_2SO_4.

Procedure:

1. Wash the burette, pipette and conical flask with water.
2. Rinse the burette with given $KMnO_4$ solution and then fill it. Remove air bubble from the nozzle and avoid leakage if any.
3. Adjust the level of $KMnO_4$ solution in burette up to zero mark with upper meniscus and fix it on a burette stand.
4. Rinse the pipette with standard 0.1 M oxalic acid solution.
5. Pipette out 10 mL of 0.1M oxalic acid solution and transfer into clean conical flask. Then add one test tube 2M sulphuric acid. (The solution remains colourless)
6. Heat the conical flask up to 60 to 70^{0}C on wire gauze or water bath (as reaction at room temperature is very slow)
7. Place the conical flask with hot solution on a white porcelain tile below the burette.
8. Start adding $KMnO_4$ solution drop wise from burette in a conical flask with continuous stirring till light/faint pink colour is obtained in conical flask. Keep the flask constant shaking in a circular manner. This is called swirling of solution. See that the colour does not disappear even on vigorous shaking. This is end point of titration.
9. Note this reading as 'Pilot reading' (pilot reading is always in whole number.)
10. Now fill the burette again with $KMnO_4$ solution up to the zero mark with upper meniscus.
11. Repeat the above procedure and take minimum three more correct burette readings.
12. Note down the constant burette reading ....(x) mL (C.B.R.)
13. From C.B.R. calculate the molarity of given $KMnO_4$ solution.

Observations:

1. Solution in a burette : $KMnO_4$
2. Solution by a pipette : oxalic acid ($H_2C_2O_4$)
3. Solution in conical flask : oxalic acid + full test tube of dil H_2SO_4
4. Indicator : $KMnO_4$ acts as self indicator
5. End point : colourless to faint pink
6. Chemical Equation : $2KMnO_4 + 5H_2C_2O_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 10CO_2\uparrow$

Observation Table:

Burette level	Pilot reading	Burette reading in mL			C.B.R.
		I	II	III	
Final	10.2	10.0	10.0	10.0	10.0 (x) mL
Initial	to	0.0	0.0	0.0	
Difference	mL	10.0	10.0	10.0	

Calculation: From above chemical equation

$$2KMnO_4 + 5H_2C_2O_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 10CO_2$$

2 moles 5 moles

2×158 g (316 g) $\equiv$ 1000 mL 5 M (molar mass of $KMnO_4$ = 158 g/mol)

$\therefore$ 1000 mL 5 M oxalic acid $\equiv$ 316 g of $KMnO_4$

$$\therefore \text{ 10 mL 0.1 M oxalic acid} = \frac{316 \times 10 \times 0.1}{1000 \times 5}$$

$$= 0.0632 \text{ g of } KMnO_4$$

$\therefore$ Hence .....(x) C.B.R. mL of $KMnO_4$ solution contains = 0.0632 g of $KMnO_4$

$$\therefore \text{ 1000 mL of } KMnO_4 \text{ contains} = \frac{0.0632 \times 1000}{10.0\text{ (x) CBR}}$$

$$\text{Hence molarity of } KMnO_4 \text{ in the solution is} = \frac{0.0632 \times 1000}{10.0\text{ (x) CBR} \times 158}$$

$$= \frac{0.4}{10.0\text{ (x) CBR}}$$

$$= 0.04 \text{ M}$$

Space for log calculation

Number	log

Result: The molarity of $KMnO_4$ solution is 0.04 M

Remark and sign of teacher: ....

MCQ

Select the correct answer for the following multiple choice type questions.

1. The molar mass of oxalic acid is
   - a. 126 u
   - ✓ **b. 126 g/mol**
   - c. 12.6 u
   - d. 12.6 g/mol
2. To prepare 1 M solution of oxalic acid
   - a. 126 g of oxalic acid dissolve in distilled water and diluted to 1 L
   - ✓ **b. 12.6 g** of oxalic acid dissolve in distilled water and diluted to 1 L
   - c. 1.26 g of oxalic acid dissolve in distilled water and diluted to 1 L
   - d. 0.126 g of oxalic acid dissolve in distilled water and diluted to 1 L
3. Water of crystallization present in oxalic acid is/are.........
   - a. 1
   - ✓ **b. 2**
   - c. 3
   - d. 5
4. Oxalic acid is used to prepare standard solution because it is a..........
   - a. substance
   - ✓ b. primary standard substance
   - c. secondary standard substance
   - d. tertiary standard substance
5. The quantity of oxalic acid required to prepare 0.1 M 100 mL standard solution of oxalic acid is
   - a. 126 g
   - b. 12.6 g
   - ✓ **c. 1.26 g**
   - d. 0.126 g

Short answer questions

1. Calculate the molar mass of oxalic acid. (H = 1, C = 12, O = 16)

Ans. Molar mass of oxalic acid ($H_2C_2O_4 \cdot 2H_2O$) is 126 g/mol.

$(2\times1) + (2\times12) + (4\times16) + (2\times(2\times1+16)) = 126$

2. Why heating is required in oxalic acid and potassium permanganate titration?

Ans. Oxalic acid is heated before titration with $KMnO_4$ solution because this reaction happens only in certain temperature.

3. Why one test tube of 2M sulphuric acid is required in permanganate titration?

Ans. Dilute sulphuric acid is added because it is stable toward oxidation and helps to generate nascent oxygen for oxidation.

4. What is the oxidation state of carbon atom in oxalic acid after completion of redox titration?

Ans. Oxidation number of carbon in oxalic acid ($C_2O_4^{2-}$) changes from +3 to +4 during titration.

5. Why primary standard substances are used to make standard solutions?

Ans. Primary standard substances are used to make standard solutions because they are pure, stable, non-hygroscopic and have a Known molar mass, so that accurate concentration can be prepared.

Remark and sign of teacher: ....